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STUDY AND RESEARCH COURSES AS AN EPISTEMOLOGICAL MODEL FOR DIDACTICS

Carl Winsløw[1]*, Yves Matheron*[2] *and Alain Mercier*[2]

1. Introduction

Contemporary educational researchers and practitioners seem to agree on the basic principle that teaching should not just *transmit* knowledge to students but should also *promote autonomous student work*, as a condition for students to learn well (whatever should be learned). There is no shortage of general frameworks proposing more elaborate characteristics of the ways in which students could work autonomously, sometimes in terms of taxonomies or scales that describe different degrees of autonomy or quality of the students' work. More general ideas, like "problem based learning" or "inquiry based science education" (with science usually defined so as to include mathematics), are increasingly adopted and prescribed by policy makers, based on the recommendation of educational researchers, such as the following:

> Inquiry-based science education (IBSE) has proved its efficacy at both primary and secondary levels in increasing children's and students' interest and attainment levels while at the same time stimulating teacher motivation. IBSE is effective with all kinds of students from the weakest to the most able and is fully compatible with the ambition of excellence. (European Commission, 2007, p. 2)

The reification achieved by assigning an acronym is noteworthy here: IBSE is not described but merely designated as if it were a unique object, obviously before our eyes. These few lines imply that IBSE is the answer to all problems in science education and that it only remains to implement it, a task in which the European Commission is currently investing millions of euro annually (as recommended by the report cited above). Even larger investments in similar programs are made by the National Science Foundation in the USA.

However, despite the consensus on the importance and benefits of student autonomy – whether individual or collective – it is also apparent that autonomy does not suffice (cf. Ben-Zvi and Sfard 2007). In fact, we believe the following to be relatively evident and consensual conditions for most institutionalized education (at least in science, including mathematics) :

(1) The goals of education are not just to acquire certain modes of work and thought. In reality, *specific "pieces" of knowledge* are included in what the student should acquire and it is usually also important for student learning that these "pieces" are structurally related – both in their cultural use and in the way they are taught and learned.
(2) Human societies cannot afford to discard the knowledge accumulated by previous generations, or to require that every generation reconstruct it anew. So, even at the level of "modes of work and thought", students should not only learn to attack problems in a "bare-

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handed way"; they should also learn to *search for* knowledge in books and other resources, and *make use of* this knowledge.

Therefore, to study and support actual teaching in real institutions, researchers need models that allow them simultaneously to describe and evaluate the quality and autonomy of students' work, and its relations to institutionalized practices, knowledge and works (within one or more disciplines). Moreover, these models must enable researchers to follow the development of student practice and knowledge over time, in particular to bring out the *dynamics* between autonomous and directed work and between creative "problem solving" and receptive study of "official knowledge". In view of these requirements, we propose in this paper a type of model which is based on the idea of *study and research course* introduced by Chevallard (2004, 2005) together with a diagrammatic "tree-like" representation of the model. We also explain how the model can be used to analyse a number of previous design studies explicitly based on the idea of SRCs, as well as a research studies of "ordinary teaching practice"; and how the model makes us refocus on the meaning of the "study" part of a SRC, and its significance for the "research" part.

2. Theoretical background on SRCs and SRC-diagrams

The theoretical construct of *study and research course* (SRC) was introduced by Y. Chevallard (2004, 2005). It was originally motivated by the introduction around the same time of a new work format in French upper secondary school – the so-called *guided personal work* format – in which students should work autonomously on transdisciplinary problems. At a more general level, the aim is to put the important and meaningful questions at the forefront of teaching, and thus depart from what Freudenthal (1973) called "anti-didactic inversion" (p. 103). This general aim is shared with many schools and methods of didactic design, including large parts of design based research in Dutch, French and German didactics since the early 1970s (see, e.g., Wittmann, 1998). In this section, we will present more specific features of the SRC construct, including the theoretical context on which it draws.

As the name says, an SRC emphasizes the dialectics of "research" (inquiry, problem solving,…) and "study" (consulting *existing* knowledge) that is in fact characteristic of *any* learning activity, even though the proportion and quality of the two elements may vary. The term "course" emphasizes that this dialectics unfolds in time and that the knowledge constructed in the process of research and study has to be analyzed in a way that captures this emergence. An SRC refers to the course of the students' study and research based on one or more questions posed by the teacher; it can also refer to the design and *a priori* analysis of such questions.

Open and targeted SRCs

Most of the examples of SRC design that have been developed in the Anglophone literature so far, explore the potential of the format as a "radical" model for teaching mathematics and other subjects, with the specific aim of enhancing the social relevance of the subjects taught and the coherence across disciplines (cf. example 1 and 2 below). In such designs, the initial questions (e.g., "What is a tsunami?") are very open and challenging, could lead in several directions and draw on more than one discipline.

However, several researchers, including Chevallard (2009), also work on SRC in mono-disciplinary settings limited to purely mathematical questions and answers (*monodisciplinary* SRCs). Much of

this work concerns SRCs that require the study and research of *certain disciplinary themes.* These *targeted* SRCs are much less "open" in terms of the derived questions and answers that students could get to work with, and they tend to be more directed by the teacher who knows the "targets". In fact, national programmes – loaded with mandatory topics to be taught and assessed – make such designs more useful and realistic in many countries (including France).

Targeted SRC still aim for students to engage in a collaborative effort of research and study, based on the knowledge they share. At the level of classroom dynamics, these targeted SRCs are similar to the teaching designs developed in the framework of the theory of didactic situations (cf. Brousseau, 1997), in that crucial elements of the new knowledge should be constructed by students as answers to *explicit questions*, which are absent in standard transmissive teaching. But the SRC format has another advantage, as even targeted and purely mathematical SRCs may start out from explicit "large" questions which are far from being tractable in one lesson. In these cases, the large question generates numerous partial answers, corresponding to derived "smaller" questions. As shown by example 3 below, such a generating question, whose study is directed by the teacher over one or more school years, may provide a coherent organization of SRCs that target large sections of the curriculum. This approach breaks away from mathematics teaching as a succession of more or less independent "chapters" where at most small and insignificant questions are raised.

Study and research courses : basic moves

The fundamental dialectics between questions and answers is at the root of the idea of a SRC. The development of knowledge by an individual or a group of individuals can be modeled in general as resulting from work with a question *Q* in the sense outlined above where the work involves one or more of the following moves, which may be highly dependent on each other:

1. Identifying "official" knowledge that can help with answering *Q*. This is the activity of "study" of *Q*, based on the *consultation of resources in media* (books, Internet and so on) and on knowledge previously studied, which can act both as part of the shared praxeological equipment of the students and as media that are (re)consulted;
2. Creating and justifying answers to *Q* through more or less *pure reasoning*. This is the activity that we designate here as "research" on *Q*; in the last section we return to the difference between this sense of "research" and the activity of university researchers.
3. Elaborating new questions from *Q*. These questions may be
    a. *subquestions* Q_1, Q_2, … for which answers provide partial answers to *Q*;
    b. *derived questions* *Q**, motivated by *Q* or by answers to *Q*, but where an answer to *Q** does not in itself answer *Q* wholly or in part.

Note that the three moves are almost always connected to one another:

- The elaboration of new questions often arises from answers of some sort;
- It is necessary to complement the identification of answers in different media with critical reasoning, in order to assess their quality and appropriateness in relation to *Q*;
- The creation and justification of answers is often supported or inspired by actual or previous consultation of "official" knowledge.

Let us now consider a simple example given by Chevallard (2004) in one of the first presentations of the idea of an SRC.

The wide availability of calculators makes it important for teachers and students alike to consider the general question *Q* of "when can one trust the calculator when it provides the same answer for two different calculations, and conclude that the two quantities calculated are really identical?". As stated, the question does not lend itself to a simple technique; but it contains challenging special cases, some of which can be indeed be associated with techniques. Chevallard treats the following case:

> Q_1: Can we trust the calculator to compare $\sqrt{a}$ and $b\sqrt{c}$ when *a, b* and *c* are positive integers less than 100 (i.e., will it give different values if $\sqrt{a}$ and $b\sqrt{c}$ are different)?

Chevallard proves – most certainly as a result of a move of type 2 – an affirmative answer A_1. In fact, two decimals suffice because $|\sqrt{a} - b\sqrt{c}| = |a-b^2c| / |\sqrt{a} + b\sqrt{c}|$ where $|\sqrt{a} + b\sqrt{c}|$ is at most 1010; if the left hand side is not 0, the difference will be visible at or before the second decimal. As a consequence of this answer, one might pose a derived question Q_1*, such as: what conditions on the size of *a*, *b* and *c* ensure that the comparison of $\sqrt{a}$ and $b\sqrt{c}$ can be done in a certain way, by just looking at the outputs produced by a standard calculator (say, capable of displaying up to 8 decimals)? On the other hand, an information search may lead to alternative and direct answers to *Q*. Googling "calculator value of square roots" leads quickly to the netforum "Dr. Math" where the response reads that "one of the problems with using calculators when doing square roots is that you don't get exact answers". After demonstrating that calculators do not give exact values when asked for the square root of a non-perfect natural number, the "math doctor" concludes:

> …besides being more exact, working with numbers in radical form is much tidier and easier than getting involved with all those messy rounded decimal approximations. (DrMath, 2004)

Chevallard also pointed out that a common "answer" to Q_1 and more generally to *Q* seems to be : "you never know with those messy calculators – do without them if you can". Indeed, a general answer may be useless without being "incorrect" in a strict sense. The SRC sketched here could be represented as a linear sequence: $Q \rightarrow Q_1 \rightarrow A_1 \rightarrow Q_1$*, with Dr. Math's answer being a possible obstacle – or alternative, or motivation – to pose Q_1.

Unlike the way questions and answers are bound to unfold in time, the essential structure of a SRC is usually not linear. Instead, the pattern of questions and answers that could arise from a first generating question *Q* is typically more like a "tree". We now elaborate this idea as a model or form of representation that was first used in this context by Hansen & Winsløw (2010) and that will serve as a main tool of analysis in the sequel.

SRC diagrams as trajectories

The example given above illustrates that, for a given question, several answers may be available and of some interest – and also that new questions may be of as much interest as the answers provided. In general the work with a question *Q* can produce a set of answers $\{A_1, A_2, \ldots\}$, a set of new questions $Q_1, Q_2, \ldots$, possibly with answers, new partial or derived questions, and so on. We can illustrate this with a tree diagram like Fig. 1, where the arrow indicates the progression; for example, the subquestion Q_3 of *Q* gives rise to one derived question and two subquestions. The shading in Fig. 1 indicates planned or observed authorship, where black questions and answers are given by the teacher, white ones by students, and grey ones are done in some sort of interaction or collaboration. In some situations, such as observed courses of study and research, one may also want to indicate the order in which questions and answers were posed (eg. by numbering the arrows etc.). In relatively teacher directed settings, this ordering may also be part of the planning of a SRC.

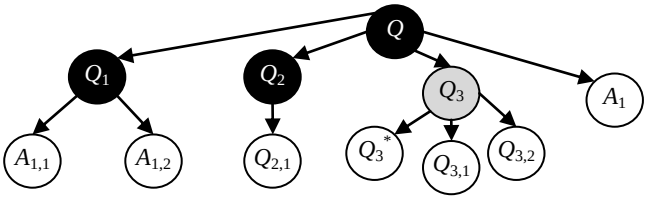


Fig. 1: Example of a SRC diagram

Of course, depending on purposes one can imagine many additions and variations to these "tree diagram" representations. For instance, analyzing observations from a realized SRC, the form of the arrows could be made to reflect whether the move is achieved mainly by study or by research. Another variation is shown in Hansen and Winsløw (2010), who color the questions according to the kinds of co-disciplinarity they represent. While additions and variations are possible, it should also be kept in mind that any form of representation entails choice and omissions. The tree diagrams as used here stress mainly the dialectics and semantic structure of questions and answers, at the expense, for instance, of the way these actually unfold in time (for such a diagrammatic representation of discourse, see Sfard and Kieran, 2001).

Before moving on, we will say a little more about the machinery provided by the Anthropological Theory of Didactics (ATD) to analyze the nature of answers and their relation among each other and with "official" knowledge that could be, or is, mobilized in the SRC.

SRC diagrams and praxeologies

In ATD, practice and knowledge are modeled as *praxeologies* consisting of four elements: a type of task (something specific to be achieved, such as drawing a circle), a technique (a way to solve the task), a technology (a way to describe and explain the technique) and a theory (a description and justification of the technology). These notions were introduced by Chevallard (1999) and are explained in more detail eg. by Barbé, Bosch, Espinoza, and Gascon (2005; see in particular section 2). Notice that questions and tasks are quite different notions here. A question, unlike a task, is always *explicit* and so it depends on a theoretical (or at least technological) environment. In addition, there are many interesting questions that cannot be "answered" by appealing to a technique.

One can consider the "answers" developed in a SRC as praxeologies, in the sense that questions motivate the learner to identify the question, or parts of it, with one or more types of tasks for which a technique – new or already known to the learner – may be applied. In the example about the degree to which calculators may be trusted, consideration of the special case (evaluation of $|\sqrt{a} - b\sqrt{c}|$) is certainly not independent of the possibility of applying the algebraic technique explained above. Similarly, teachers ask questions of students with the aim that they should find the answers in praxeologies to be learned or already learned. The questions themselves may have to be developed into subquestions or derived questions in order to link them to praxeologies, and particularly technologies, with which students are already familiar, and to those they are about to learn. The responsibility for organizing and adapting the *questions* worked on by the students is often assumed entirely by the teacher. It is the teacher who introduces new technology because the teacher controls the questions and often also the media with which the students are working. This situation allows the teacher to ensure a certain coherence in students' work within a class and over a period of time. But it also tends to reduce the activity of students to working with very small questions, given by the teacher and always with the answers in the close neighborhood of what has

just been worked on. In ATD terms, such a "theme" of praxeologies is bound together by a single technology, such as computation with square roots.

Teachers' "thematic autism" (cf. Barbé et al., 2005) designates the situation in which teachers essentially plan teaching to "cover" one theme after the other, with little or no demand for students to relate the themes with each other: first we calculate with square roots, then we do circle theorems, and so on. The unsatisfactory results from this type of teaching are the main motivation to investigate the viability of letting students work with "bigger" questions, which necessitates more than local technical work, and hence more than identifying "official knowledge" which can help answering the question (move 1 above). While teachers are often bound to "cover" certain themes and sectors (in the sense of ATD, see Barbé et al., 2005), the idea is that they can do so more efficiently and coherently through SRCs departing from strategically chosen "generating questions" – with a considerable space for action left to students when it comes to develop the study into something manageable, to seek relevant material to study, and so on.

This space may be reduced for various reasons and in various ways, as in the case of a targeted SRC where certain answers are to be worked with, while still retaining the central and visible role of larger questions. For instance, instead of teaching the circumscribed circle of a triangle as an isolated curiosity, the following question will enable treatment of this and many other parts of secondary level geometry: "how many circles pass through k points (k = 1,2,3,…)". In a usual classroom, the teacher remains the director of the study process and the main "medium" of information, and the students cannot be given the same autonomy as a collective of professional researchers; still, working from and with "larger questions" is expected to increase the rationality and coherence of the study process in the classroom.

We now present in more detail some examples of how this idea has been developed in actual design research, carried out in real classes under a variety of institutional conditions.

3. SRC as model for didactic design work

In this section, we show how to synthesize the design of a SRC using the diagrammatic approach introduced above. Our first two examples illustrate a tendency for research involving SRC design to focus on the viability of *extra-mathematical questions* with strong potential to generate mathematical study and research, including in particular study and research on mathematical models of "real" phenomena. In many institutional contexts it is not possible to realize such work in view of the curriculum or the teachers' capacities, or other constraints. The third example we give comes from attempts to include SRCs in such more constrained contexts, while retaining the ambition of coherence and meaningfulness.

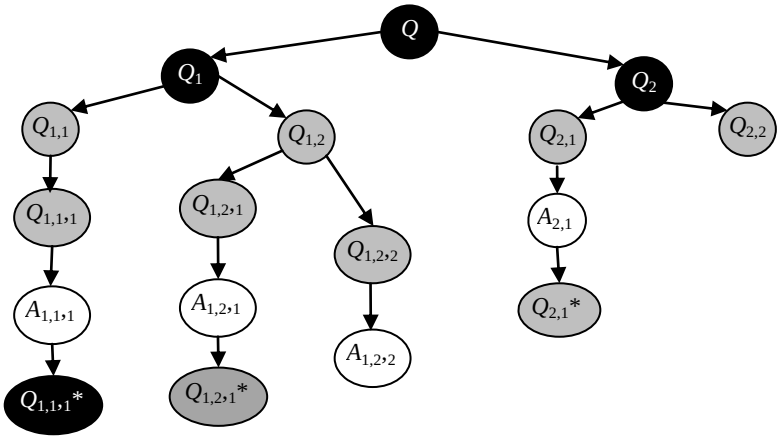
Example 1: A SRC in a Spanish university course : models of population growth

Barquero, Bosch and Gascón (2007) constructed and experimented with a SRC for engineering students, based on the following generative question:

> *Q*: From data showing the development of the size of a population of geese on an island over a period of time, how can we predict population size at a later time?

It should be emphasized that students are only given data showing the population size in five consecutive years; in principle it is up to them to construct a model. But to organize the process further, more structure is given in the design. The authors designed three SRCs corresponding to three subquestions: Q_1 corresponding to discrete models with recursive equations depending on the previous value ($Q_{1,1}$) or on several previous values ($Q_{1,2}$); Q_2 corresponding to continuous models based on differential equations. These are taken up consecutively by the whole class and as determined by the teacher, based on the ideas generated during a first period of work with the general question (Barquero et al., 2007, p. 2055). The analysis prior to the experimentation outlines a system of further subquestions and answers similar to, but at some points more detailed than, those shown in Fig. 2. The branches can be interpreted as corresponding to different sectors and themes that are to be covered by the curriculum, but now in a way "unified" by the question Q. The shading of questions and answers tries to capture the distribution of responsibility among teacher and students in the actual realization of the SRC reported on: black ones "given" by the teacher, the others arising more (white) or less (gray) from the study and research of students.

The initial analysis of the questions that could be developed from Q (cf. Fig. 2) serves to illustrate how a SRC could cover "a wide range of the mathematics taught in a first year university course" (Barquero et al., 2007, p. 2053), and do so by way of related inquiries on a common generating question. Fig. 2 also outlines what was covered in the actual realization and how: notice that some branches end with questions, such as $Q_{1,1,1}$*. In principle, the questions in a given branch of the tree arise in a logical way from previous answers or questions, with the increasing complexity as one proceeds downwards being motivated by shortcomings of answers such as "unlimited growth" found with exponential type models. However, it is also clear also from the realized SRC that the teacher remains responsible for maintaining the progression in this direction, if the previously mentioned "coverage" of material is to be assured. It is not clear if the students will be able to undertake the comparison of alternative "end" results belonging to different branches such as the Leslie models ($A_{1,2,2}$) and logistic growth (resulting from $Q_{2,1}$*) that are treated as distinct SRCs in the formulation of Barquero et al. (2007, p. 2053).



Q: You are given data showing the development of the size of a population of geese on an island, over a period of time. How can we predict its size at a later time?

Q_1: … discrete models, population size x_n given at times $t = nt_0$ where t_0 is fixed ("generation")

$Q_{1,1}$: … assuming that a generation depends only on the previous one : $x_{n+1} = f(x_n)$)

$Q_{1,1,1}$: … assuming constant relative growth: $x_{n+1} = (1 + k)x_n$

$A_{1,1,1}$: Malthusian model, $x_n = (1 + k)^n x_0$

$Q_{1,1,1}$*: The answer $A_{1,1,1}$ is unrealistic as growth is unlimited. How can assumptions be modified – what about other models of type $x_{n+1} = f(x_n)$, where *f* is a C^1-function?

$Q_{1,2}$: … assuming that the *n*th generation x_n depends on the *d* previous ones : $X_{n+1} = f(X_n)$, where

$$X_n = (x_{nd}, x_{nd+1}, \ldots, x_{nd+d-1}) \in \mathbb{R}^d$$

$Q_{1,2,1}$: …assuming linear model $X_{n+1} = AX_n$ where *A* is a $d \times d$ -matrix

$A_{1,2,1}$: After *n* periods: $X_n = A^n X_0$

$Q_{1,2,1}$*: How does A^n behave? Stability as $n \to \infty$?

$Q_{1,2,2}$: ….assuming affine model $X_{n+1} = AX_n + b$ where *A* is a $d \times d$ -matrix and $b \in \mathbb{R}^d$, what happens?

$A_{1,2,2}$: After *n* periods: $X_n = A^n X_0 + \sum_{k=0}^{n-1} A^k b$ (stable as $n \to \infty$ under some conditions)

Q_2: … continuous modeling the population size as a continuous function: $x(t)$ where *t* is time

$Q_{2,1}$: assuming constant relative growth: $x'(t) = kx(t)$

$A_{2,1}$: Exponential growth, $x(t) = be^{at}$ (in particular, unlimited growth)

$Q_{2,1}$*: … How about relative growth with upper bound: $x' = kx(K - x)$ (logistic model)

$Q_{2,2}$: How about other continuous models, like $x' = f(x,t)$, or higher order ODE models?

Fig. 2: Outline of SRC designed and experimented by Barquero et al. (2007).

The experiment was done in the context of an "optional workshop" within a mandatory first year mathematical foundations course for engineering students. Considering that calculus and linear algebra are usually main the components in such a course, it is appropriate that central results can be activated or developed from a common and relatively "real" problem situation. However, the

authors neither claim nor suggest that this SRC design could function as a complete alternative to the "main" university course, which develops as a well structured succession of "answers". In other words, the design seems to function essentially as a valuable supplement to the standard course that allows the students to deepen their relation to the praxeologies introduced in that course, and in particular to establish relations between them. In the actual realization, as reported in Barquero et al. (2007), one sees mainly alternative *answers* produced by students, such as polynomial interpolations to fit the initial data. The fact that most questions are shaded grey indicates that they are identified by students and teachers collectively, based on previous student group work, apparently with essentially all questions being prepared by the teacher. The point of this design is thus not for students to construct questions "freely" but for questions to arise "naturally" from the progression of students' work with the generating question under the guidance of the teacher. The design therefore has potential to allow for a combination of research and study activity.

Example 2: A SRC in a Danish high school : analyzing a three point shot

Thrane (2009) designed and experimented with four SRCs for third year students in Danish high school as part of a joint course on mathematics and physical education. The *a priori* analysis of one of these SRCs, entitled "Three point shots in basketball", is represented in Fig. 3.

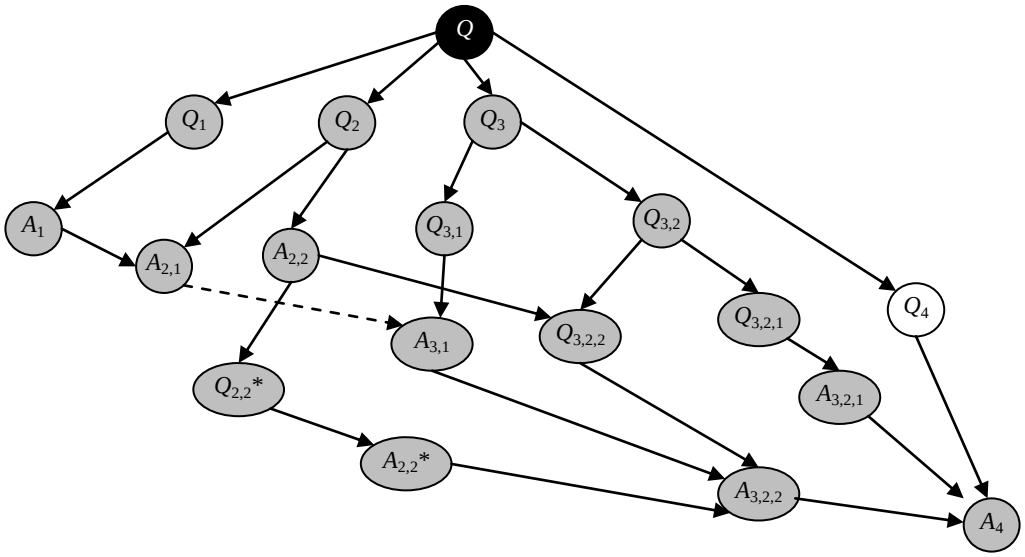
The use of SRCs in this context was motivated by a recent (2007) curriculum change for Danish upper secondary school, introducing short courses of "study preparation" in which students work with broad questions supported by two or more disciplines. In this case, such a course combined mathematics and physical education, around the theme "Modelling and designing human movements". It was organized in four SRCs, each worked on by students in groups for 1-2 hours.

The four SRCs were independent in the sense that different sports and mathematical models were involved. A common feature was the use of the software LoggerPro (cf. Wyrembech, 2009), which allows the user to produce mathematical models from video. For example, the programme can produce an approximate equation or parametrization of a curve with a video clip and a "marked point to be followed" as input.

In the SRC shown in Fig. 3, the students were organized in groups. During a physical education lesson, each group filmed the others' attempts to do three point shots in basketball, and chose one attempt to be analysed in view of improving the shot. The SRC took this video as a point of departure, together with the generating question

> *Q*: Beginning with a video clip showing a three point shot in basketball, how can LoggerPro be used to model the shot and what suggestions can be made for a training program to improve the shot?

The question is not a very "big" one, like that in Example 1; it resembles an exercise in two parts, while there is still room for generating work using a variety of approaches and levels of sophistication.



Q: Beginning with a video clip showing a three point shot in basketball, how can LoggerPro be used to model the shot and what suggestions can be made for a training program to improve the shot?

Q_1: Find a parametrization $f: [a, b] \to \mathbb{R}^2$ of the curve captured by LoggerPro, following the ball from when it leaves the hand of the player till it reaches the area near the basket.

A_1: Fixing the origin at the point where the ball leaves the players hand at $t = 0$, and tagging a sufficient number of points through which the ball passes, Logger Pro produces a model of type $f(t) = (ct, -at^2 + bt)$ for $t \in [0, d]$, where *a*, *b*, *c* and *d* are positive decimal numbers, and *f*(*d*) is the position of the ball as it hits the plate.

Q_2: How can the velocity of the ball at a point along the curve be found?

$A_{2,1}$: By differentiating the result in $A_{1,1}$ to get $f'(t) = (c, -2at + b)$

$A_{2,2}$: Using the same procedure as in $A_{1,1}$ but plotting instead the direction of the ball at a point, we get in principle $f'(t)$; however it appears that we get a small negative first order term in $f_1'(t)$, corresponding to a small horizontal deceleration ("backspin"). [The difference from $A_{2,1}$ can be explained by technical assets of Logger Pro.]

$Q_{2,2}$*: How does the shot depend on the "backspin" of the ball, which seems to cause the ball to "brake" in the vertical direction?

$A_{2,2}$*: With unchanged vertical acceleration (gravitation) $f_2''(t)$, it results in a smaller value of $f_1'(t) / f_2'(t)$.

Q_3: If the ball hits the plate behind the basket, how could the shot be changed so that the ball falls more "vertically" towards the ring (so that while it is reflected by the plate, it still hits within the ring)?

$Q_{3,1}$: What is the angle α between the incoming ball's curve and the (vertical) plate behind the curve?

$A_{3,1}$: As $f'(d)$ is tangent vector to the curve at the point where the ball hits the plate, $\alpha = \arctan(f_1'(d) / f_2'(d))$.

$Q_{3,2}$: How does α depend on the shot?

$Q_{3,2,1}$: Let β be the angle between vertical and the direction of the shot. How does α depend on β ?

$A_{3,2,1}$: From the geometry of the parabola, we see that larger values of α go with larger values of β. (This could be made more precise using $A_{2,1,1}$ and the corresponding computation of β, while it would only refer to the actual shot).

$Q_{3,2,2}$: How does α depend on the "backspin" ? (cf. $A_{2,2}$)

$A_{3,2,2}$: From $A_{3,1,1}$ and $A_{2,2}$*, increasing the "backspin" would make α decrease.

Q_4: What should the training focus on?

A_4: From $A_{3,2,1}$ and $A_{3,2,2}$, the angle α could be decreased (resulting in a better chance of "goal") by decreasing β (more "vertical" shot) and increasing the "backspin".

Fig. 3: Outline of SRC designed and experimented by Thrane (2009).

There is clearly a study part of the activity, which relies heavily on material such as text books that the students are using in the two disciplines and on previous experience with the software. The research part relies on, and is driven by, the confrontation of models with data and the students' experience from the physical exercise. In practice, these circumstances give rise to a rather large variation of student trajectories, including "subquestions" different from those in Fig. 3 and also a few irrelevant questions and "false" answers.

As an example, one group accomplished the path Q_1-A_1-Q_2-$A_{2,1}$ essentially as foreseen in Fig 3, used the velocity function only to compute the initial angle and speed of the shot, and then jumped to Q_4 saying simply that "it's a nice shot but it lacks precision". Probably without a time constraint they might have advanced on this last question. Another group only used LoggerPro to verify that the ball follows a parabola, and then worked through the branch Q_3- $Q_{3,2}$ - $Q_{3,2,1}$- $A_{3,2,1}$ to conclude with the corresponding part of A_4 ("try to shoot more vertically"). While no group got anywhere near the complete SRC of Fig. 3, all except $A_{2,2}$ and the parts depending on it (i.e., concerning the effect of "backspin") were realized by some group and shared with the rest of the class in a "presentation session". It seemed reasonable for Thrane (2009) to conjecture that with two or three 2-hour sessions, the whole SRC in Fig. 3 could be worked through by a high school class with similar preparation in analytic geometry up to parametrizations of 2-dimensional curves and some experience with LoggerPro. The teacher would still have to point out "missing" questions after the first session, or in this case an alternative method for Q_2 leading ultimately to $Q_{2,2}$*.

These considerations point to at least two main functions of an analysis like that shown in Fig. 3. Firstly, while the diagram itself is always a preliminary hypothesis about what can be developed from the question, the design can be developed systematically based on experience from observations. For example, it appeared from Thrane (2009) that although Q_1 appeared among students because they had recently studied parametrizations, they would need more time and experience to relate the answers to the rest of the SRC. The second function is more practical and concerns the usefulness of the analysis for actual teaching: it allows the teacher to plan ahead for relating different approaches to the main question and, while directing the course, to decide whether and how a substantial continuation of the work could be "prompted" by strategic input from the teacher. While the teacher would still have to assess more locally the correctness and strength of students' work, the diagram enables a more global view to be maintained.

Example 3. An SRC on relative numbers in French lower secondary school

Our third example of SRC design, represented in Fig. 4, comes from the work of (CD)AMPERES, a major French network of 80 secondary school teachers and five researchers, established with the explicit aim of developing and testing SRCs in lower and upper secondary school. A number of documents from the project, all in French, can be found in Matheron (2010).

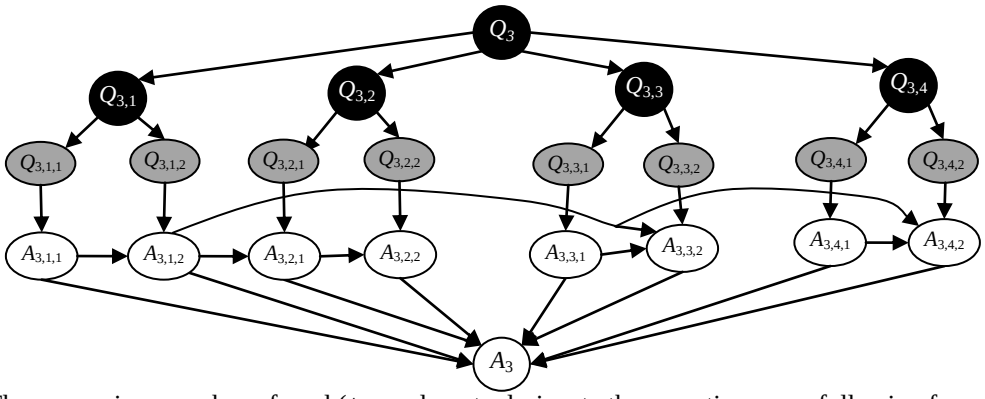
The example is related to the French curriculum for two successive grade levels with students aged between 12 and 14, and it concerns the introduction of relative numbers (sometimes the theme is referred to as negative or directed numbers). The curriculum is very explicit as regards the techniques to be taught to add, subtract, compare and multiply relative numbers. However, it is

silent about the technological and theoretical levels of these praxeologies: for example, it gives no indication of how to explain the existence of relative numbers, or how to justify the rule $-(-a) = a$.

The text books used by most teachers suggest an introduction to relative numbers based on cultural phenomena such as temperature, time scales, debt, and profit rather than on mathematical constructions. In the text books, these metaphors are prolonged to yield justification of facts related to order, addition, and subtraction. It is well known that this approach does not work for multiplication (see Hefendehl-Hebeker, 1991): instead the current text books propose "extending the pattern" of the usual multiplication table. Thus, nothing remains of the structural ("new math") approach introduced in the 1970s. But many French teachers and students are unsatisfied with the metaphorical approach, and for students it leads to difficulties as the different metaphors seem to reinforce or create certain epistemological obstacles to the appropriation of relative numbers (cf. Hebeker, 1991 for classical examples).

The SRC on relative numbers was constructed by one of the (CD)AMPERES teams. It has been tested in ordinary classrooms by the teachers in the team and stretched over 15 lessons distributed across two years. The sessions were filmed and analyzed, which led to many detailed observations that we do not report on here. What we want to explain is the overall structure of the SRC, as illustrated in Fig. 4; the text part of this figure also gives the critical questions and answers of the study and research carried out by teachers and students.

The SRC was constructed following an idea proposed by Chevallard. The generating question Q_3 emerges from a preparatory calculation activity (cf. Fig. 4), where students are asked to calculate, mentally and quickly, the value of $a + b - c$, where a, b and c are positive numbers, and $c - b \leq a$. At first, they only calculate from left to right. But with suggestive examples, such as 435 + 61 – 62, they notice that subtracting 1 from 435 appears to work. The teacher asks for a justification for this strategy. Students search and find, either by considering the algorithm for subtraction or more intuitively ("take one point from 435 to give it to 61", as one students says) students find that they can justify the technique by first transforming 435 to 434+1. With more examples, students become familiar with expressions such as "adding 61 and subtracting 62 from a number is the same as subtracting 1". Expressions like "subtracting 1" are called "calculation programs", and after a while the teacher suggests denoting those corresponding to subtraction of a number as –1, –3.4, and so forth.



Q_3 : The expressions you have found (+*m* and –*n*, to designate the operations on *a* following from calculations of type $a + b - c$, where *a*, *b*, *c* are "ordinary decimal numbers") can they be considered as numbers? Can you do with them what you do with ordinary numbers – if so how?
$Q_{3,1}$: … can you add them ?
$Q_{3,1,1}$: If so, what would be the sum of +7 and +2?
$A_{3,1,1}$: In fact, as adding 7 and then adding 2 (or in fact, adding 2 then 7) corresponds to add 9, we can add these types as we add the corresponding numbers. We may thus identify +*m* with *m*.
$Q_{3,1,2}$: Then how about the other ones, like the sum of +7 and –2, or the sum of –7 and –2 ?
$A_{3,1,2}$: As +7 = 5+2, and as adding 2 and then subtracting 2 is the same as adding 0, the sum of +7 and –2 is the sum 5+0 = 5 = 7–2; so adding –2 is just subtracting 2; by a similar reasoning we get that the sum of –7 and –2 is –9.
$Q_{3,2}$: … can you subtract them ?
$Q_{3,2,1}$: If so, what would be the difference of +7 and +2 ?
$A_{3,2,1}$: Identifying +*m* with *m* we get 7–2 = 5.
$Q_{3,2,2}$: Then how about the other ones, like the difference of +7 and –2, or the difference of –7 and –2 ?
$A_{3,2,2}$: We get (for instance) $-7-(-2) = -7+2+(-2)-(-2) = -7+2 = -5$ by $A_{3,1,2}$.
$Q_{3,3}$: …can you compare (order) them?
$Q_{,3,3,1}$: How about comparing numbers of type +*m*, like +7 and +2?
$A_{3,3,1}$: As we identify +*m* with the ordinary number *m*, we order them as usual: 7 > 2.
$Q_{,3,3,2}$: How about comparing numbers like –7 and +2, or –7 and –2?
$A_{,3,3,2}$: Extending the additive property of order, as $-7+8 = 1 < 6 = -2+8$ (using here $A_{3,1,2}$), we get –7 < –2, etc.
$Q_{3,4}$: …can you multiply them?
$Q_{,3,4,1}$: How about multiplying numbers of type +*m*, like +7 and +2?
$A_{3,4,1}$: As we identify +*m* with the ordinary number *m*, we have the usual values: $(+7)\cdot(+2) = 7\cdot 2 = 14$.
$Q_{3,4,2}$: How about multiplying numbers like –7 and +2, or –7 and –2?
$A_{3,4,2}$: Extending the distributive property of the product, we get, using $A_{3,1,2}$ and $A_{3,4,1}$: $(-7)\cdot 2 + 7\cdot 2 = ((-7)+7)\cdot 2 = 0\cdot 2 = 0$, so that $(-7)\cdot 2 = -(7\cdot 2) = -14$ (by $A_{3,4,1}$), and from this follows $(-14) + (-7)\cdot(-2) = (-7)\cdot(2+(-2)) = 0$ whence $(-7)\cdot(-2) = 14 = 7\cdot 2$.
A_3: Yes, in fact we may add, subtract, order and multiply these numbers, as developed above.

Fig. 4: The main part of an SRC on negative number arithmetic

Students now work with this new notation, and find rules such as –5+4–2 = –3 ("subtracting 5, then adding 4, then subtracting 2, is to subtract 3") etc. This leads to the crucial question Q_3 (see Fig. 4). The teacher and students develop the subquestions corresponding to different things one can do with "ordinary" numbers, and the teacher organizes the students' research on these questions. For instance, a list is made of calculation programs that could be constructed by subtraction and addition of two given positive numbers. In a class where these numbers were 7 and 2, some students found that both –7 + (–2) and –7 – (–2) are the same as –9. This result seems to violate previously established principles for calculating with numbers, which suggest that the sum and difference of

two numbers are not the same. This outcome leads to the questions of how to perform addition and subtraction for the new numbers in a mathematically valid way. Sometimes, external media were mobilized. One student asked an older brother and came back to class with the following hypothesis: "calculate with two negatives as one positive". The teacher would come back to this hypothesis once a valid rule had been established for the addition. For instance, to calculate 7+ (–2), some students propose to "break up 7" to get 7 = 5 + 2 + (–2), which gives 5. This method enables the teacher to pose a new question: why would 2 + (–2) be 0? In this way, the SRC follows a dialectic of students' search for answers (sometimes supported by the teacher as "media") and emerging new questions, formulated by students or the teacher. Although the four questions $Q_{3,1}$, …, $Q_{3,4}$ each organize about 3-5 hours of teaching and are essentially posed by the teacher, the treatment of subquestions (like $Q_{3,1,1}$) thus proceeds in a much more variable way, depending on the students' work.

Notice that $Q_{3,4}$ and the corresponding branch of the SRC are, treated one year after the rest, according to the curriculum. It is not uncommon for SRCs designed in the CD(AMPERES) project to stretch over more than one school year.

4. SRC as modeling tool in analyzing didactic processes

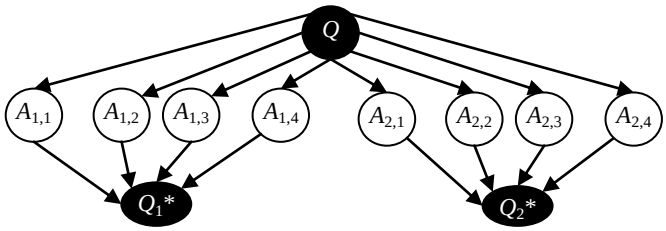
A main claim of this paper is that diagrams of the type illustrated above can be used to study *any didactic process*, including teaching conceived without the idea of SRC in mind. Two examples which concern individual project work in Danish high school, were presented by Hansen and Winsløw (2010). Here, we present another example of how more or less "ordinary" teaching can be analyzed and understood in terms of the structure of the questions and answers they involve, as captured by the diagram.

Example 4. An open approach lesson in Japan.

Miyakawa and Winsløw (2009) analysed a 6th grade lesson on similarity and proportion as a typical example of teaching design in the Japanese "open approach" tradition (cf. also Nohda, 1991). In Fig. 5 is the diagrammatic analysis of this lesson, for which we had access to source data such as the lesson plan and a video recording. It is a simple structure and different from the previous examples.

In the lesson, the teacher first poses the question Q for individual work and carefully observes the answers developed by students during the individual work. The answers in Fig. 5 are essentially those that the teacher has foreseen, while a few more appear in practice: they can be said to arise from an individual "research" process as no external works can be consulted. The main part of the lesson is taken up by the successive presentation of student answers, orchestrated by the teacher. Positive answers $A_{1,k}$ are presented before negative answers $A_{2,k}$ – a process which in a sense combines study of others' arguments with research to produce refutation and other reactions. The teacher refrains completely from validating or invalidating answers, and closes the lesson by posing the two derived questions Q_1* and Q_2* – to be answered by those who gave a positive and a negative answer, respectively, to Q. In this way, the following lesson will continue to develop different answers to the more general question: *How may one define the meaning of saying that two figures are "of the same form"*? Because this general question has not been raised and certainly not answered, the teacher does not expect or aim for the question Q to be definitively answered in the

lesson. Indeed, this situation reflects a property of questions commonly asked outside mathematics: the answer may depend on how you interpret the terms. The general question on how one may define "the same form" needs to be raised by students, while the questions Q, Q_1* and Q_2* proposed by the teacher could guide students in that direction.



Q: Consider two rectangles, one measuring 3cm×5cm and another measuring 5cm×7cm. Are they of the same form or not – and why?
$A_{1,1}$: Yes, because they have the same angles.
$A_{1,2}$: Yes, because the difference between breadth and height is the same for the two triangles ($b_1–h_1= b_2–h_2$)
$A_{1,3}$: Yes, because the breadths and heights differ by the same ($b_1–h_1= b_2–h_2$).
$A_{1,4}$: Yes, because the larger rectangle is obtained by adding the same length to both sides of the smaller rectangle
$A_{2,1}$: No, because the angle between the diagonal and the base is different for the two rectangles.
$A_{2,2}$: No, because the ratio between height and breadth is different for the two triangles ($b_1/h_1 \neq b_2/h_2$).
$A_{2,3}$: No, because the ratio between the heights is different from the ratio of the breadths ($b_1/b_2 \neq h_1/h_2$).
$A_{2,4}$: No, because if we magnify the small rectangle to have height 5cm, the breadth becomes 5cm*5/3 = 25cm/3 which is larger than 7cm.
Q_1*: Draw a rectangle with a different form from the smaller rectangle (3cm×5cm).
Q_2*: Draw a rectangle with the same form as the smaller rectangle (3cm×5cm).

Fig. 5: SRC diagram to analyse a Japanese "open approach" lesson.

The lesson is part of a course on proportion and ratio, so it is clearly a long term goal to arrive at a definition of "same form" that would favor negative answers to Q. But to bring out the central and non-trivial character of this definition it is worth going along with other conceptions such as that reflected by $A_{1,1}$, which are useful in some situations but may become problematic after a while (as in the context of Q_1*).

Note also in this and many other mathematics lessons, there may be "implicit generating questions". In this case the generating question could be, "how can we give precise mathematical meaning to the notion of 'same form', and how can we decide, in a precise and easy way, if two given figures are of the same form?". However, such a general question is not explicit in the lesson, and not even in the lesson plan, which focuses on the notions of ratio and proportion as required by the programme. One strength of SRC design, therefore, exists in the requirement to identify and pursuing generating questions, even in strongly regulated contexts such as that considered in Example 3.

5. Perspectives

The four cases considered in the two preceding sections represent a heterogeneous collection of didactic processes, all of which were analyzed as SRCs based on the web of questions and answers potentially arising (Section 3) or actually realized (Section 4). Clearly this type of analysis focuses

attention on the negotiation of knowledge and can only be carried out if rather detailed information is available about the contents, order and sources of questions and answers.

We noted in Section 4 that such an analysis is sometimes feasible even for research done without the agenda of studying or creating SRCs. What matters is access to enough details on the dynamics of questions and answers raised in the process of exchanging and diffusing knowledge. The model proposed in this paper applies to these didactical phenomena, and in particular the diagrams serve to give a structural representation of how knowledge is constructed in the process. But the model could also be more than a tool for research. We hypothesize that such a representation is sufficiently close to teachers' concerns, and also capture such essential parts of a didactic design, that one could use it as a tool for collaboration and communication with and among teachers, regarding concrete teaching designs. While the actual staging of a SRC always depends on a number of evident local conditions, such as time and resources available and students' experience with this and similar activities, the diagrams represent a more stable analysis of the generating question in terms of derived and partial questions and the answers to which they could lead. In this paper, this hypothesis is to some extent supported by the design of example 2, which is the work of a high school teacher who tested it, as part of his master thesis work. Similar designs are currently being developed, using the same representational tool, by other Danish high school teachers. It is too early to state anything firm about the potential roles of the tool in the wider dissemination of such designs.

Diagrams of designs aimed at producing SRCs (such as Fig. 2-4 here) tend to have numerous and long branches, as they contain a generating question from which a chain of subquestions could arise. It is interesting to notice that even in these cases, most questions are in black, indicating that they are asked, or to be asked, by the teacher, and those that are grey, where authorship is in principle open, are also likely to be asked by the teacher. For the teacher, the need to keep moving through a course is often linked to ensuring the minimum of "good questions" is asked, and teachers are not often comfortable to leave this responsibility to the students.

Using any model in design or in analysis focuses our attention on specific aspects of the situation. We have already mentioned the focus of the SRC model on students' inquiry and autonomy, which is shared by many other didactic models. With the SRC model, we are led to pay specific attention to the students' participation and autonomy in the crucial part of inquiry that consists of formulating and organizing the questions on which the inquiry rests, and we have noted that this part is often limited, even in courses deliberately designed as SRCs. This observation gives room for the hypothesis that in previous SRC designs, and certainly in studies of "ordinary" teaching practices, the conditions for enabling substantial *study activities* have been given too little attention.

To advance this hypothesis we need to reconsider what could be meant by study in this context. A weak form of study occurs when students solve end-of-chapter exercises from a text book and look up ideas or results in the chapter or in the preceding parts of the book. This activity is not too different from the weak form of research that amounts to recalling and reproducing methods from recently studied material, and applying them to a problem at hand. One could say that these kinds of study and research are interior to the institution in the sense that all sources are constrained by methods and materials from previous, typically recent, teaching. Of course, challenging problems for "research" may still be realized in such a setting: but the "study" part is limited and entirely controlled by the institution, represented by a teacher. This is essentially the case for all of the

examples studied above. Mathematics may not be the only school discipline in which "exterior" media, such as books or journals, Internet, and television, are rarely integrated into institutionalized study – but the rarity is amplified by the perceived irrelevance of exterior media to the aims of the discipline, or at least by the perceived difficulty to make relevant use of these media. The Internet, in particular, seems to offer an interesting and worthwhile resource for many mathematical questions, although it is evidently a resource that one must learn to use critically – as shown by the example of Dr. Math, cited above. However, we believe this to be true for any authentic study activity.

To sum up, the relative absence of genuine study in the examples considered in sections 3 and 4 (and in many other published examples) seems to be related to the relatively strong control of the question part of the processes we analyzed. This situation corresponds perfectly with various forms and arrangements of "organized research", with students working on challenging questions formulated by the teacher in order to acquire certain knowledge. In other perhaps more common settings, the research part is also missing in the sense that students are only given the option to work with narrow questions and only when the teacher has provided the students with all elements needed to answer.

As an "extreme" departure from this setting, there are SRCs based on just one generating question, no material particularly designed for students' study activity, and with "everything" – including all of the Internet – available as a resource for students autonomously to advance the inquiry and, in particular, to develop the questions. While such a situation is neither realistic nor desirable in institutional teaching, retaining student autonomy with respect to study and study material represents a challenge to mathematics teachers. Indeed, the necessity of discriminating and evaluating the quality of "answers" found in media that are not warranted by the institution could be a potentially rich source of student inquiry in mathematics. This challenge needs to be met by each teacher individually to the extent that students are allowed to search for and choose study material, for instance by using a search engine. The potentially negative effect of such an approach on the teacher's control of questions and answers in the didactic process needs to be considered further, including the pursuit of "non-relevant" questions and the difficulty for the teachers and students alike to make critical use of sources found by chance.

While these risks are neither to be denied nor diminished, a significant contribution of the SRC model is to prompt reflection on the meaning of research and study in a didactic process and on the possibilities arising from their interaction. In much teaching, research and study tend to occur more or less separately. They are also separate in the cases examined in Sections 3 and 4: on the one hand, there are research activities in so-called "progressive", "inquiry oriented" lessons, and on the other hand, students study in privacy. However, it could be argued that students should learn how to make use of public and easily accessible resources found on the Internet as materials to study for the purpose of finding the answer to a question – even in mathematics. One obvious advantage of this activity for the purpose of *research* is that *the answers found are not a priori validated by the institution*, and so there is a stronger motivation for asking students to assess them critically. When it comes to students asking new questions, the integration of study and research activities could also be rich. Even if the study of a certain materials does not lead to an answer for the motivating question, it could support the formulation of subquestions that students can answer, derived

questions to explore, new materials to study, and so on. However, school mathematics rarely allows students to experience study beyond "looking up the full answer", with the possibilities of failure and success, but nothing in between.

Of course, one cannot expect students and school children to reproduce the complete integration of study and research found in the work of professional mathematicians or in other theoretical subjects. On the other hand, the SRC model provides us with a means for analyzing and designing teaching that goes beyond the "pure research" ideals that are sometimes named as "inquiry based teaching", "problem solving" and so on, because it additionally incorporates of the study dimension of professional research. More practically, the model emphasizes the need to integrate the study dimension into research like teaching activities, so that students and teachers could at least occasionally try to and draw on the full and unrestricted range of mathematical resources that can be found outside of text books. It must be recognized that such an integration is not often realised and that we do also lack stable didactical engineering that integrates these resources, to achieve more complete research and study courses.

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